Meeting ?? SP23

SmartNICs

@crawfish

Outline

Networking

The NIC

The SmartNIC

Some Cool Extensions

Section 1

Networking

Networking

- A computer network is a set of computers sharing resources located on or provided by network nodes
  - Peer-to-peer
  - The internet
- Network communication can be very fast
  - Wireless connections are high latency
  - Data-centers use fast wired connections
- Modern networking is built on a 5-layer 'stack'

Network Stack

1. Application
2. Transport
3. Network
4. Data Link
5. Physical

Network Stack

1. Application
   - HTTP, SMTP, etc.
   - Generates data, requests connections
2. Transport
3. Network
4. Data Link
5. Physical

Network Stack

1. Application
2. Transport
   - TCP/UDP
   - Establishes connections with remote host
   - Keeps track of running processes
3. Network
4. Data Link
5. Physical

Network Stack

1. Application
2. Transport
3. Network
   - IP
   - Creates packets that move accross network
   - Uses (virtual) IP addresses to identify source, destination
4. Data Link
5. Physical

Network Stack

1. Application
2. Transport
3. Network
4. Data Link
   - Ethernet, Wi-Fi, PPP
   - Creates network frames which encapsulate packets
   - Uses (physical) MAC addresses to identify source, destination
5. Physical

Network Stack

1. Application
2. Transport
3. Network
4. Data Link
5. Physical
   - Physical cables, ports, etc.
   - Where actual bits are transmitted

Network Stack

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Section 2

The NIC

Background

- Network Interface Card/Controller
- Appeared at Xerox PARC in 1973

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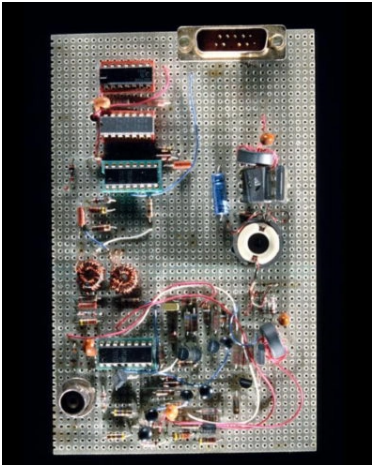
- Network Interface Card/Controller
- Appeared at Xerox PARC in 1973
  - Ethernet
- Eventually became industry standard
- Now NIC $\approx$ computer networking
  - *Technically* USB network adapters are not NICs

Xerox PARC NIC



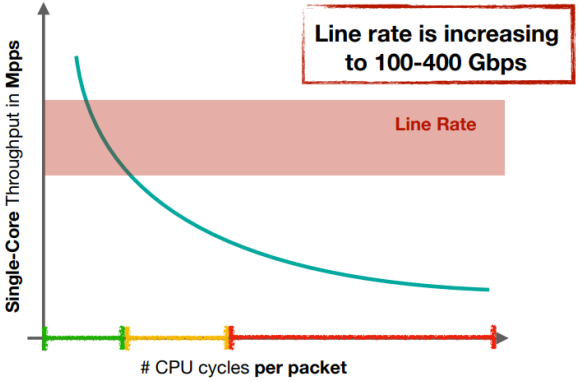
The NIC

- Hardware interface between the computer and the network
  - Used for both wired + wireless networks
  - Allows devices to talk to one another
- Any network packet will go through the NIC
  - Encapsulates bits into packets and vice versa
  - Connected to CPU via PCIe

The NIC

- Physical layer is fairly simple + unchanging
  - Great place for hardware functionality
- In comparison, application, transport, and network layers can be very complicated
  - Also change frequently
- In the past, this division of labor has worked well
  - CPU was able to process packets as fast / faster than it recieved them

Software Packet Processing is Slow



Section 3

The SmartNIC

Why Smart?

- SmartNICs are smarter NICs
  - Network interface cards with more functionality than foundational NICs

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  - Minimize data movement
  - Reduce the main CPU's workload

Why Smart?

- SmartNICs are smarter NICs
  - Network interface cards with more functionality than foundational NICs
  - Minimize data movement
  - Reduce the main CPU's workload
- Used to accelerate networking, storage, security functionality
  - Can also provide virtualization / routing functions

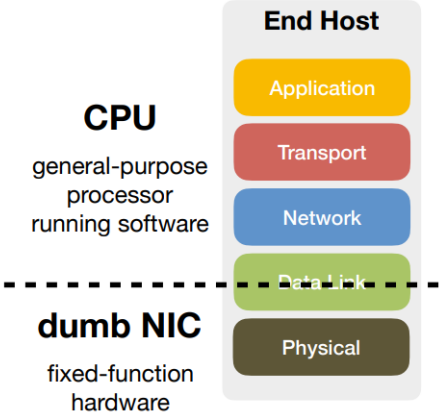
SmartNICs

- Programmable
  - Users can pick which parts of packet processing to offload
  - Account for rapidly changing, widely varied requirements

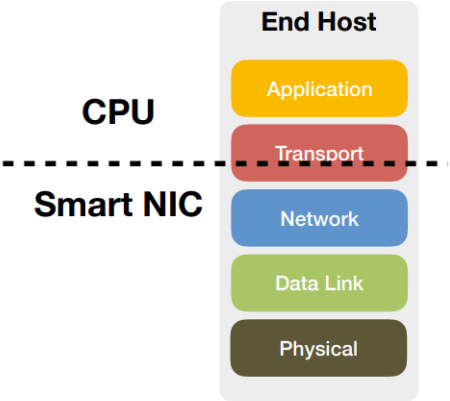
SmartNICs

- Programmable
  - Users can pick which parts of packet processing to offload
  - Account for rapidly changing, widely varied requirements
- Domain-specific
  - Hardware is optimized for network processing

Fundamental NIC



SmartNIC


End Host
CPU
Application
Transport
Smart NIC
Network
Data Link
Physical

Hardware

- FPGAs
  - Hardware resources are highly customizable
  - Easy / cheap to change
  - Good for highly parallelizable computations
  - Higher throughput + lower latency = better performance
    - At the cost of programability

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- SoC
  - System of weak cores + memory hierarchy
  - Programming model is closer to SW
  - Architecture can be specialized for network processing
  - Slower than FPGA based options, but easier to work with

Section 4

Some Cool Extensions

FlowBlaze

- One major problem w/ NICs is programability
  - Can be more difficult to write performant code for smartNICs (especially FPGA based)
- Providing a good abstraction (FlowBlaze) can help alleviate this
  - Expressive, performant framework
- Adapts 'match-action tables' to describe the evolution of network's **flow state** using FSMs
  - Match-action table → List of actions to perform on packets, executed based on packet headers

FlexNIC

- HW I/O virtualization can eliminate hypervisor / OS overhead
- Prevents some other useful techniques
  - Memory overcommit
- FlexNIC allows applications / OSes to install rules into the NIC
  - Will execute on packets while exchanging them with host memory
  - Can include how data is exchanged, where portions are placed
- Integrates well with software-defined networking trends

The end.